

Fitting and Interpreting Exponential Models

Answer Key

Precalculus

Quick Answers

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|---|---|
| 1. (a) balance at $t = 0$, in dollars = 1200, (b) annual percent increase = 4.5 | after 12 years, in dollars = 12591 |
| 2. (a) hourly growth factor $b = 2$, (b) predicted cell count at $t = 7$ hours = 25600 | 7. 15.1 hours |
| 3. (a) percent lost each hour = 14, (b) milligrams left after 6 hours = 101.1 mg | 8. (a) annual percent decrease = 9, (b) actual percent decrease over 10 years = 61.1, — |
| 4. 14.45 years | 9. (a) single yearly growth factor form = 3001.1236^t , (b) annual percent growth rate = 12.36, — |
| 5. (a) yearly decay factor $b = 0.8$, (b) value after 5 years, in dollars = 7864 | 10. (a) temperature when poured = 83 degrees, (b) minutes until 45 degrees = 15.3 minutes |
| 6. (a) continuous rate $k = 0.077$, (b) balance | |
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What is verified

18 of 20 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 8, 9. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSF-BF.B.5 – *problems 6, 9*

HSF-LE.A.4 – *problems 1, 2, 3, 4, 5, 7, 8, 10*

Difficulty 1–5 of 5 across 20 tagged checks.

1. A savings account is modelled by $A(t) = 1200(1.045)^t$ dollars, t in years. (a) Balance at $t = 0$. (b) Annual percent rate of increase.

(a) In $A = ab^t$ the coefficient a is the value at $t = 0$, because $b^0 = 1$: $A(0) = 1200(1.045)^0 = 1200$. 1200

(b) The base is the yearly multiplier, so the yearly percent change is $b - 1$ written as a percent: $1.045 - 1 = 0.045 = 4.5\%$ increase. 4.5%

2. A culture holds 200 cells at the start and 3200 cells 4 hours later, modelled by $N(t) = ab^t$. (a) Find b . (b) Predict $N(7)$.

(a) Two points determine a and b . The first gives $a = 200$. The second gives

$$200b^4 = 3200 \implies b^4 = \frac{3200}{200} = 16 \implies b = 16^{1/4}.$$

Since $2^4 = 16$, the hourly growth factor is $b = 2$: the culture doubles every hour.

$b = 2$

(b) With $N(t) = 200 \cdot 2^t$, at $t = 7$: $N(7) = 200 \cdot 2^7 = 200 \cdot 128 = 25600$ cells.

25600

3. $M(t) = 250(0.86)^t$ mg, t in hours. (a) Percent lost each hour. (b) Amount left after 6 hours, to the nearest 0.1 mg.

(a) Each hour the amount is multiplied by 0.86, so 14% of what is present is lost:

$$1 - 0.86 = 0.14.$$

14%

(b) Substitute $t = 6$ and evaluate once, at the end:

$$M(6) = 250(0.86)^6 = 250(0.4045672\dots) = 101.1418\dots$$

Rounded to the nearest tenth of a milligram.

101.1 mg

Common wrong answer: 40 mg, from treating the loss as linear — subtracting 14% of the original 250 mg six times. Decay is repeated multiplication, not repeated subtraction.

4. $P(t) = 18000(1.023)^t$ people. Find, to the nearest hundredth of a year, when $P = 25000$.

Solution: isolate the power, then take logarithms of both sides — the unknown is in the exponent, which is exactly what a logarithm undoes.

$$\begin{aligned} 18000(1.023)^t &= 25000 \\ (1.023)^t &= \frac{25000}{18000} = 1.3888\dots \\ t \ln(1.023) &= \ln\left(\frac{25000}{18000}\right) \\ t &= \frac{\ln(25000/18000)}{\ln(1.023)} = \frac{0.328504\dots}{0.022739\dots} = 14.4464\dots \end{aligned}$$

Rounded to the nearest hundredth of a year.

$t \approx 14.45$ years

Common wrong answer: 1.03, from writing $\ln(25000)/\ln(18000)$. The quotient of two logarithms is not the logarithm of the quotient.

5. A car is worth 24000 dollars new and 15360 dollars after two years, modelled by $V(t) = ab^t$. (a) Find b . (b) Value after 5 years.

(a) Here $a = 24000$, and the second data point gives

$$24000b^2 = 15360 \implies b^2 = \frac{15360}{24000} = 0.64 \implies b = \sqrt{0.64}.$$

Take the positive root, since a value factor must be positive.

$b = 0.8$

(b) $V(5) = 24000(0.8)^5 = 24000(0.32768) = 7864.32$, so to the nearest dollar the car is worth 7864

6. $A(t) = 5000(1.08)^t$ dollars is rewritten as $A(t) = 5000e^{kt}$. (a) Find k to four decimal places. (b) Find the balance after 12 years.

(a) The two forms agree for every t exactly when the bases match: $e^k = 1.08$. Taking natural logarithms, $k = \ln(1.08) = 0.0769610\dots$ $k \approx 0.0770$

(b) Keep the exact $k = \ln(1.08)$ inside the calculation and round only at the end:

$$A(12) = 5000e^{12\ln(1.08)} = 5000(1.08)^{12} = 5000(2.5181701\dots) = 12590.85\dots$$

To the nearest dollar.

12591

7. $N(t) = 40(0.5)^{t/6}$ mg, t in hours. How long until 7 mg remains, to the nearest 0.1 hour?

Solution: the exponent is $t/6$, not t — the tracer halves once every 6 hours.

$$\begin{aligned} 40(0.5)^{t/6} &= 7 \\ (0.5)^{t/6} &= \frac{7}{40} = 0.175 \\ \frac{t}{6} \ln(0.5) &= \ln\left(\frac{7}{40}\right) \\ t &= 6 \frac{\ln(7/40)}{\ln(0.5)} = 6(2.514575\dots) = 15.0874\dots \end{aligned}$$

Rounded to the nearest tenth of an hour.

$t \approx 15.1$ hours

Common wrong answer: 2.5 hours, from dropping the 6 in the exponent and solving as if the tracer halved every hour.

8. $W(t) = c(0.91)^t$ million litres. (a) Annual percent decrease. (b) Actual percent decrease over 10 years. (c) Explain why "9 percent a year for 10 years" is not a 90 percent drop.

(a) $1 - 0.91 = 0.09$, so use falls by 9% of the current amount each year.

9%

(b) After ten years the multiplier is $(0.91)^{10} = 0.3894161\dots$, so the fraction remaining is about 0.3894 and the fraction lost is

$$1 - (0.91)^{10} = 0.610584\dots = 61.0584\dots\%$$

To the nearest tenth of a percent.

61.1%

(c) *Instructor-judged.* A correct response says that the 9% is taken from a *shrinking* base each year, so the yearly decreases multiply rather than add: the model multiplies by 0.91 ten times, which leaves about 38.9% and so removes about 61.1%, not 90%. Adding 9% ten times would only be correct if the same fixed number of litres were removed every year. Award full credit only when the multiplicative reason is given, not the number alone.

9. Analyst A: $F(t) = 300(1.06)^{2t}$. Analyst B: $F(t) = 300(1.1236)^t$. (a) Rewrite A with one yearly factor. (b) B's annual percent growth. (c) Justify that the models agree for every t .

(a) Use the power law $x^{mn} = (x^m)^n$ on the exponent $2t$:

$$300(1.06)^{2t} = 300((1.06)^2)^t = 300(1.1236)^t,$$

since $(1.06)^2 = 1.1236$ exactly.

$$F(t) = 300(1.1236)^t$$

(b) The yearly factor is 1.1236, so the annual rate is $1.1236 - 1 = 0.1236$.

$$12.36\%$$

(c) *Instructor-judged.* A correct justification cites the law of exponents, not a single test value: raising 1.06 to the power $2t$ is the same as raising $(1.06)^2$ to the power t , and $(1.06)^2$ is exactly 1.1236, so the two expressions are equal for every t . Checking one year (for example $t = 1$) shows agreement at that year only and earns partial credit.

10. $T(t) = 21 + 62(0.94)^t$ degrees Celsius, t in minutes; the room stays at 21 degrees. (a) Temperature when poured. (b) Time to cool to 45 degrees.

(a) At $t = 0$ the power is 1, so $T(0) = 21 + 62(1) = 83$. The 62 is the initial gap above room temperature, not the temperature itself.

$$83 \text{ degrees}$$

(b) Subtract the room temperature first — only the gap decays exponentially:

$$21 + 62(0.94)^t = 45$$

$$62(0.94)^t = 45 - 21 = 24$$

$$(0.94)^t = \frac{24}{62} = 0.387096\dots$$

$$t = \frac{\ln(24/62)}{\ln(0.94)} = \frac{-0.949081\dots}{-0.061875\dots} = 15.3386\dots$$

Rounded to the nearest tenth of a minute.

$$t \approx 15.3 \text{ minutes}$$

Common wrong answer: 5.2 minutes, from forgetting the 21 degree room temperature and solving $62(0.94)^t = 45$.